

Magnetoelastic wave in Ferromagnetic Thin-film Mediated by Dipolar Interaction: Supplementary data

Hiroki Yoshida¹, Ryohei Kono¹, Manato Fujimoto², Motoki Asano³,
Daiki Hatanaka³, Kei Yamamoto^{4,5}, Shuichi Murakami^{2,5,6}

¹ *Department of Physics, Institute of Science Tokyo, 2-12-1 Ookayama, Meguro-ku, Tokyo 152-8551, Japan*

² *Department of Applied Physics, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan*

³ *Basic Research Laboratories, NTT, Inc., 3-1 Morinosato-Wakamiya, Atsugi-shi, Kanagawa 243-0198, Japan*

⁴ *Advanced Science Research Center, Japan Atomic Energy Agency, Tokai, Ibaraki 319-1195, Japan*

⁵ *Center for Emergent Matter Science, RIKEN, 2-1 Hirosawa, Wako, Saitama 351-0198, Japan*

⁶ *International Institute for Sustainability with Knotted Chiral Meta Matter (WPI-SKCM²), Hiroshima University, Higashi-hiroshima, Hiroshima 739-0046, Japan*

1 Jacobi matrix

Here we calculate the Jacobi matrix and its determinant of the transformation $\mathbf{r} \mapsto \mathbf{R} = \mathbf{r} + \mathbf{u}(\mathbf{r}, t)$ up to first order of $\mathbf{u}$.

The definition of the Jacobi matrix J is

$$J_{ij}(\mathbf{r}) := \delta_{ij} + \frac{\partial u_i}{\partial r_j}. \quad (1)$$

Then, its determinant (Jacobian) is given as

$$\begin{aligned} \det J &= \varepsilon_{i_1 i_2 i_3} J_{1i_1} J_{2i_2} J_{3i_3} \\ &= \varepsilon_{i_1 i_2 i_3} \left(\delta_{1i_1} + \frac{\partial u_1}{\partial r_{i_1}} \right) \left(\delta_{2i_2} + \frac{\partial u_2}{\partial r_{i_2}} \right) \left(\delta_{3i_3} + \frac{\partial u_3}{\partial r_{i_3}} \right) \\ &= 1 + \frac{\partial u_1}{\partial r_1} + \frac{\partial u_2}{\partial r_2} + \frac{\partial u_3}{\partial r_3} + \mathcal{O}((\partial u)^2) \\ &\approx 1 + \nabla \cdot \mathbf{u}. \end{aligned} \quad (2)$$

For a regular three by three matrix A , its inverse matrix is given by

$$A^{-1} = \frac{1}{\det A} C^T, \quad (3)$$

where C is a cofactor matrix defined as

$$C_{ij} := \frac{1}{2} \varepsilon_{ii_1 i_2} \varepsilon_{jj_1 j_2} A_{i_1 j_1} A_{i_2 j_2}. \quad (4)$$

Then, the inverse of the Jacobi matrix is

$$\begin{aligned}
\left[(J^{-1})^T\right]_{ij} &= (J^{-1})_{ji} \\
&= \frac{1}{\det J} C_{ij} \\
&= \frac{1}{2 \det J} \varepsilon_{ii_1 i_2} \varepsilon_{jj_1 j_2} J_{i_1 j_1} J_{i_2 j_2} \\
&= \frac{1}{2 \det J} \varepsilon_{ii_1 i_2} \varepsilon_{jj_1 j_2} \left(\delta_{i_1 j_1} + \frac{\partial u_{i_1}}{\partial r_{j_1}} \right) \left(\delta_{i_2 j_2} + \frac{\partial u_{i_2}}{\partial r_{j_2}} \right) \\
&\approx \frac{1}{2} (1 - \nabla \cdot \mathbf{u}) \left(\varepsilon_{ii_1 i_2} \varepsilon_{jj_1 j_2} + \varepsilon_{ij_1 i_2} \varepsilon_{jj_1 j_2} \frac{u_{i_2}}{r_{j_2}} + \varepsilon_{ii_1 j_2} \varepsilon_{jj_1 j_2} \frac{u_{i_1}}{r_{j_1}} \right) \\
&= \left(1 - \frac{\partial u_{i_1}}{\partial r_{i_1}} \right) \delta_{ij} + \frac{1}{2} \left(1 - \frac{\partial u_{i_1}}{\partial r_{i_1}} \right) \left(2 \delta_{ij} \frac{\partial u_{i_1}}{\partial r_{i_1}} - 2 \frac{\partial u_j}{\partial r_i} \right) \\
&\approx \delta_{ij} - \frac{\partial u_j}{\partial r_i}.
\end{aligned} \tag{5}$$

2 Derivation of the dipolar field

Here we derive the dipolar field that cause the magnetoelastic coupling.

First, Maxwell's equations in the coordinate $\mathbf{R}$ are

$$\nabla_{\mathbf{R}} \cdot (\mathbf{H}_{\text{in}}(\mathbf{R}, t) + \mathbf{M}(\mathbf{R}, t)) = 0, \tag{6}$$

$$\nabla_{\mathbf{R}} \times \mathbf{H}_{\text{in}}(\mathbf{R}, t) \approx 0, \tag{7}$$

where $\nabla_{\mathbf{R}}$ is a derivative with respect to $\mathbf{R}$. We go back to the coordinate $\mathbf{r}$ from these equations. Here we need to pay attention to the conservation of the magnetization [1]:

$$\begin{aligned}
\int_{V_R} d^3 R \mathbf{M}(\mathbf{R}, t) &= \int_{V_r} d^3 r \mathbf{M}(\mathbf{r}, t) \\
\Rightarrow \int_{V_r} d^3 r (\det J) \mathbf{M}(\mathbf{R}(\mathbf{r}), t) &= \int_{V_r} d^3 r \mathbf{M}(\mathbf{r}, t) \quad \forall V_r. \\
\therefore \mathbf{M}(\mathbf{R}) &= \frac{1}{\det J(\mathbf{r})} \mathbf{M}(\mathbf{r}).
\end{aligned} \tag{8}$$

Then, using the Jacobi matrix of the coordinate transformation, the Maxwell's equations become

$$\begin{aligned}
&\nabla_{\mathbf{R}} \cdot (\mathbf{H}_{\text{in}}(\mathbf{R}, t) + \mathbf{M}(\mathbf{R}, t)) = 0 \\
\Rightarrow \left[(J^{-1})^T\right]_{ij} \frac{\partial}{\partial r_j} \left(h_i(\mathbf{r}, t) + \frac{1}{\det J} M_i(\mathbf{r}) \right) &= 0,
\end{aligned} \tag{9}$$

and

$$\begin{aligned}
&\nabla_{\mathbf{R}} \times \mathbf{H}_{\text{in}}(\mathbf{R}, t) \approx 0 \\
\Rightarrow \varepsilon_{ijk} \left[(J^{-1})^T\right]_{il} \frac{\partial}{\partial r_l} h_j(\mathbf{r}) &\approx 0 \quad \forall k = x, y, z.
\end{aligned} \tag{10}$$

From the results of the previous section 1, up to the first order of $\mathbf{m}$ and $\mathbf{u}$, we get

$$\nabla_r \cdot (\mathbf{h}^{\text{in}}(\mathbf{r}) + \mathbf{m}(\mathbf{r})) = \mathbf{M}_0 \cdot \nabla_r (\nabla_r \cdot \mathbf{u}(\mathbf{r})), \tag{11}$$

$$\nabla_r \times \mathbf{h}^{\text{in}}(\mathbf{r}) = 0, \tag{12}$$

for points inside the thin film. The equations outside of the film is the same as the previous section:

$$\begin{aligned}
&\nabla \cdot \mathbf{h}^{\text{out}}(\mathbf{r}, t) = 0, \\
&\nabla \times \mathbf{h}^{\text{out}}(\mathbf{r}, t) \approx 0.
\end{aligned}$$

Because $\mathbf{h}^{\text{in/out}}$ are rotation-free even with magnetoelastic coupling under the magnetostatic approximation, it is possible to introduce a scalar potential as

$$\mathbf{h}^{\text{in/out}} = -\nabla\phi^{\text{in/out}}.$$

Outside of the film, the solution is given by

$$\phi^{\text{out}} = \begin{cases} Ae^{+kz} & (z < -\frac{L}{2}) \\ Be^{-kz} & (z > +\frac{L}{2}), \end{cases} \quad (13)$$

because $\phi^{\text{out}} \rightarrow 0$ at $|z| \rightarrow \infty$.

The boundary conditions are given in a form of continuity of magnetic field parallel to the surface and the continuity of magnet flux density normal to the surface:

$$\mathbf{H}^{\text{in}} \cdot \mathbf{t} = \mathbf{H}^{\text{out}} \cdot \mathbf{t}, \quad (14)$$

$$(\mathbf{H}^{\text{in}} + \mathbf{M}) \cdot \mathbf{n} = \mathbf{H}^{\text{out}} \cdot \mathbf{n}, \quad (15)$$

where $\mathbf{t}$ and $\mathbf{n}$ are tangential and normal vectors of the surface, respectively. Since the system is uniform in the y direction, we consider in the xz -plane. In this case, these vectors are given by

$$\mathbf{t} = \left(1 + \frac{\partial u_x}{\partial x}, 0, \frac{\partial u_z}{\partial x}\right), \quad (16)$$

$$\mathbf{n} = \left(-\frac{\partial u_z}{\partial x}, 0, 1 + \frac{\partial u_x}{\partial x}\right). \quad (17)$$

Using the form of the solution (13), we get the modified boundary condition for ϕ^{in} as

$$(\pm k + \partial_z) \phi^{\text{in}}|_{z=\pm\frac{L}{2}} = (m_z - ikM_{0,x}u_z)|_{z=\pm\frac{L}{2}}, \quad (18)$$

where signs correspond to each other. Thus, we need to solve a differential equation

$$\nabla^2 \phi^{\text{in}} = \nabla \cdot \mathbf{m} - \mathbf{M}_0 \cdot \nabla (\nabla \cdot \mathbf{u}(\mathbf{r})), \quad (19)$$

under the boundary condition (18). For this purpose, we introduce a Green function

$$\nabla^2 g(z, z') = \delta(z - z'), \quad \left(z, z' \in \left(-\frac{L}{2}, \frac{L}{2}\right)\right), \quad (20)$$

with a boundary condition

$$(\pm k + \partial_z) g(z, z')|_{z=\pm\frac{L}{2}} = 0. \quad (21)$$

Using this $g(z, z')$, ϕ^{in} is written as

$$\phi^{\text{in}}(z) = \int dz' [\nabla g(z, z') \cdot \mathbf{m}(z') - \nabla (\mathbf{M}_0 \cdot \nabla g(z, z')) \cdot \mathbf{u}(z')]. \quad (22)$$

By a standard procedure using Fourier transformation, we can solve Eqs. (20) under (21) as

$$g(z, z') = -\frac{1}{2k} e^{-k|z-z'|}. \quad (23)$$

By taking the gradient, we get the dipolar field in the main text as

$$\mathbf{h}^{\text{in}}(z) = \int_{-\frac{L}{2}}^{\frac{L}{2}} dz' [G^m(z, z') \mathbf{m}(z') + G^u(z, z') \mathbf{u}(z')], \quad (24)$$

where the first term is the same as in Ref. [2] and

$$G^m(z, z') = \begin{pmatrix} -G_P(z, z') & 0 & -iG_Q(z, z') \\ 0 & 0 & 0 \\ -iG_Q(z, z') & 0 & G_{P'}(z, z') \end{pmatrix}, \quad (25)$$

$$G^u(z, z') = -ikM_{0,x}G^m(z, z'), \quad (26)$$

$$G_P(z, z') = \frac{k}{2} e^{-k|z-z'|}, \quad (27)$$

$$G_Q(z, z') = \text{sgn}(z - z') G_P(z, z'), \quad (28)$$

$$G_{P'}(z, z') = G_P(z, z') - \delta(z - z'). \quad (29)$$

3 Derivation of the force

Here we calculate the force induced by the magnetoelastic coupling.

The Euler–Lagrange equation about u_i^* gives

$$\int dz \Re \left[\rho \ddot{u}_i - \left(-\mu_0 \frac{\delta h_a^*}{\delta u_i^*} h_a + \partial_i \sigma_{ij} \right) \right] = 0, \quad (30)$$

where the second term corresponds to the volume force caused by the dipolar field. By straightforward calculations, we can show that

$$\int_{-\infty}^{\infty} dz'' (G^m(z'', z))^* G^m(z'', z') = \int_{-\infty}^{\infty} dz'' G^m(z, z'') G^m(z'', z') = -G^m(z, z'), \quad (31)$$

$$\int_{-\infty}^{\infty} dz'' (G^m(z'', z))^* G^u(z'', z') = \int_{-\infty}^{\infty} dz'' G^m(z, z'') G^u(z'', z') = -G^u(z, z'). \quad (32)$$

Then,

$$\begin{aligned} \int dz \frac{\delta h_a^*(z)}{\delta u_i^*} h_a(z) &= \int dz dz' dz'' (+ikM_{0,x}) (G^m(z, z'))_{ia}^* [G^m(z, z'') \mathbf{m}(z'') + G^u(z, z'') \mathbf{u}(z'')]_a \\ &= -ikM_{0,x} \int dz' \int dz'' [G^m(z', z'') \mathbf{m}(z'') + G^u(z', z'') \mathbf{u}(z'')]_i \\ &= -ikM_{0,x} \int dz' h_i^{\text{in}}(z'). \end{aligned} \quad (33)$$

Thus, we get

$$f_i(z) = -ikM_{0,x} h_i(z). \quad (34)$$

References

- [1] Kei Yamamoto and Sadamichi Maekawa. Magnetostatic Field Induced by Mechanical Deformations. *Ann. Phys. (Berlin)*, 536:2300395, 2023.
- [2] B. A. Kalinikos and A. N. Slavin. Theory of dipole-exchange spin wave spectrum for ferromagnetic films with mixed exchange boundary conditions. *J. Phys. C: Solid State Phys.*, 19:7013, 1986.